

CAPSTONE PROJECT PROPOSAL

AJIO E-Commerce Automation Testing Using Playwright

Project Information

Project Title: AJIO E-Commerce Automation Testing Using Playwright

Application: AJIO (<https://www.ajio.com>)

Prepared By: Arun N

Automation Tool: Playwright

Programming Language: JavaScript

Framework Design: Page Object Model (POM)

Project Type: End-to-End Web Automation Testing

Total Planned Test Cases: 120+

OBJECTIVE

The primary objective of this project is to design and implement a scalable enterprise-level automation testing framework for the AJIO e-commerce platform using Playwright.

The framework focuses on validating critical user workflows including authentication, product search, filtering, cart functionality, checkout process, user profile management, responsive UI behaviour, and API validation.

The project also emphasizes cross-browser execution, reusable framework architecture, stable automation, reporting, debugging, and enterprise-level testing practices.

Technology Stack

Category	Tools/Technologies
Automation Framework	Playwright
Language	JavaScript
Runtime Environment	Node.js
Framework Design	POM

IDE	VS Code
Reporting	HTML & Allure
Version Control	Git & GitHub
CI/CD	GitHub Actions

Core Functional Modules & Components

1. Authentication & Login Validation
 2. Homepage & Navigation Validation
 3. Product Search Functionality
 4. Product Listing & Product Detail Validation
 5. Filters & Sorting Functionality
 6. Add to Cart & Cart Validation
 7. Checkout & Shipping Address Validation
 8. API Testing & Cross-Browser Validation
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Expected Deliverables

- Enterprise-grade Playwright automation framework
 - Organized Page Object Model architecture
 - Automated reusable test suites
 - API testing implementation
 - Cross-browser execution support
 - HTML and Allure reports
 - GitHub repository with documentation
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Conclusion

This capstone project demonstrates the implementation of a scalable and maintainable enterprise-level automation testing framework using Playwright for a real-world e-commerce platform.

The framework validates critical business workflows, UI components, and end-to-end shopping functionalities while following industry-standard automation engineering practices to ensure application quality, reliability, and maintainability.